InnovizTwo for Emergency Brake Assist: derivation of the statistical model

3DSSF – Kozma, Exercise 3

Question. An object of width W, height H and reflectivity ρ is at distance R. What is the probability that the lidar gets at least $k_{\min} = 5$ returns from it in one frame, and up to which R is that probability at least 99 %?

1 Number of beams on the object

The lidar samples a regular angular grid with spacing $\Delta\theta_h$ (horizontal) and $\Delta\theta_v$ (vertical). At distance R neighbouring beams are $s = R\,\Delta\theta$ apart (small angles: $\tan\Delta\theta \approx \Delta\theta$, error $< 10^{-6}$ for $0.1°$). One side of the object therefore spans

$$L = \frac{W}{R\,\Delta\theta_h} \quad \text{beam spacings.}$$

Random grid offset. Beams sit at $x_j = u + js$, with the offset u uniform on $[0, s)$ (the object can be anywhere relative to the grid). The interval $[0, W]$ contains $\lfloor L \rfloor + 1$ beams if $u < W - \lfloor L \rfloor s$, otherwise $\lfloor L \rfloor$. Hence

$$P\big(n_h = \lfloor L \rfloor + 1\big) = \frac{W - \lfloor L \rfloor s}{s} = L - \lfloor L \rfloor = \{L\}, \qquad P\big(n_h = \lfloor L \rfloor\big) = 1 - \{L\}.$$

The same holds vertically, independently. So $n = n_h n_v$ takes 4 values with known probabilities w_{ij}. As a check, $\mathbb{E}[n_h] = L$: on average the count is exactly the geometric value.

2 Detection probability of one beam

Signal. For a Lambertian target larger than the laser spot, the lidar equation gives the received power

$$P_r \propto \frac{\rho}{R^2}\, e^{-2\alpha R},$$

where $e^{-2\alpha R}$ is the loss in the air on the way out and back (Beer–Lambert, extinction coefficient α).

From signal to probability. We model the detected signal events in one beam as Poisson with a mean λ proportional to the received power, $\lambda = C\rho R^{-2} e^{-2\alpha R}$. A beam gives a return if at least one event is detected:

$$p(R, \rho) = 1 - e^{-\lambda} = 1 - \exp\Big(-C\,\frac{\rho}{R^2}\, e^{-2\alpha R}\Big).$$

This is the simplest curve that is 0 for no signal, tends to 1 for a strong signal and has only one free constant. That matters, because the datasheet gives only one point.

Calibration. The datasheet states $p(R_0, \rho_0) = P_0$ in clear air ($\alpha = 0$):

$$1 - e^{-C\rho_0/R_0^2} = P_0 \implies C = -\ln(1 - P_0)\,\frac{R_0^2}{\rho_0}.$$

With $R_0 = 210\,\text{m}$, $\rho_0 = 0.1$, $P_0 = 0.9$ this gives $C\rho/R^2 = \ln 10 \cdot (R_0/R)^2$ for $\rho = 0.1$, so at $R = 250\,\text{m}$: $p = 1 - e^{-2.303 \cdot 0.7056} = 1 - e^{-1.625} = \mathbf{0.803}$.

3 Number of returns and the requirement

Given n beams on the object, each returning independently with probability p,

$$K \mid n \sim \text{Binomial}(n, p), \qquad P(K \geq k \mid n) = 1 - \sum_{i=0}^{k-1} \binom{n}{i} p^i (1-p)^{n-i}.$$

Averaging over the four beam counts (law of total probability):

$$P(K \geq k) = \sum_{i,j \in \{0,1\}} w_{ij}\, P(K \geq k \mid n_{ij}).$$

Verdict. $R_{99} = \max\{R : P(K \geq 5) \geq 0.99\}$; the object passes if $R_{99} \geq 250\,\text{m}$.

Worked example (dark pedestrian, $0.5 \times 1.8\,\text{m}$, $250\,\text{m}$, clear). $s_h = 0.436\,\text{m}$, $L_h = 1.15$; $s_v = 0.218\,\text{m}$, $L_v = 8.25$.

n	w	$P(K \geq 5 \mid n)$	contribution
$1 \times 8 = 8$	$0.85 \cdot 0.75 = 0.64$	0.947	0.604
$1 \times 9 = 9$	$0.85 \cdot 0.25 = 0.21$	0.982	0.209
$2 \times 8 = 16$	$0.15 \cdot 0.75 = 0.11$	1.000	0.112
$2 \times 9 = 18$	$0.15 \cdot 0.25 = 0.04$	1.000	0.037
			$\mathbf{0.962} < 0.99$

The mean is $\mathbb{E}[K] = \mathbb{E}[n]\, p = 1.15 \cdot 8.25 \cdot 0.803 = 7.6 \geq 5$: the average passes, the probability fails.

4 Weather

Weather reports give the visibility V (meteorological optical range), not α. Visibility is defined as the distance at which the contrast of a dark object drops to 2 %, so (Koschmieder)

$$e^{-\alpha V} = 0.02 \implies \alpha = \frac{-\ln 0.02}{V} = \frac{3.912}{V}.$$

For each condition c we evaluate the model with α_c. The share of driving time π_c gives the yearly figure, again by total probability:

$$P_{\text{weighted}}(K \geq 5) = \sum_c \pi_c\, P(K \geq 5 \mid \alpha_c) = 0.88\, P_{\text{clear}} + 0.08\, P_{\text{rain}} + 0.04\, P_{\text{fog}}.$$

Pedestrian at 250 m: $0.88 \cdot 0.962 + 0.08 \cdot 0.803 + 0.04 \cdot 0.047 = 0.913$.

5 Parameters and why we chose them

Parameter	Value	Reason / source
Lidar: InnovizTwo datasheet v3.1 (2024), Condor configuration, region of interest		
$\Delta\theta_h \times \Delta\theta_v$	$0.1° \times 0.05°$	datasheet, ROI resolution. The ROI ($30° \times 9.6°$) is about 130×40 m at 250 m, so the lane ahead is inside it.
$R_0,\ \rho_0,\ P_0$	210 m, 10 %, 0.90	datasheet: "210 m @ 10 % reflectivity", note 4: "True Positives = 90 % per pixel" (100 klx sunlight). This is exactly the per-beam probability the model needs.
$R_{\max}$, frame rate	300 m, 20 FPS	datasheet; results are capped at 300 m.
Requirement and scenario		
$R_{\text{req}},\ k_{\min}$	250 m, 5	given by the task.
confidence	99 %	our choice: an EBA must not miss an object in more than about 1 frame in 100. Sensitivity: 95 % gives 253 m for the pedestrian, 99.9 % gives 219 m.
ρ	10 %	the standard worst case in lidar datasheets; dark clothing and tyres are 5–10 %. 50 % (bright clothing) is shown for comparison.
pedestrian	0.5×1.8 m	adult shoulder width ≈ 0.5 m, height ≈ 1.75–1.8 m.
car	1.8×1.5 m	rear of a typical passenger car (width 1.75–1.85 m, height 1.45–1.5 m).
motorcycle	0.8×1.4 m	motorcycle with rider seen from behind (handlebar/mirrors, rider's head).
truck	2.5×3.5 m	rear of a truck/trailer; EU limits are 2.55 m wide and 4 m high.
tyre / cargo	0.5×0.5 m	lost load or a car tyre lying on the road (tyre diameter 0.6–0.7 m, seen partly).
Weather		
clear	$\alpha = 0$	datasheet conditions; visibility > 20 km gives $\alpha < 0.0002$/m, negligible at 250 m.
rain / haze	$V = 5$ km	moderate rain or haze; $\alpha = 7.8 \cdot 10^{-4}$/m.
fog	$V = 1$ km	the WMO definition of fog is visibility below 1 km, so this is the mildest fog; $\alpha = 3.9 \cdot 10^{-3}$/m. Denser fog is worse.
shares π_c	88 / 8 / 4 %	our estimate, not official: Budapest has about 80 days per year with ≥ 1 mm rain, but rain lasts hours, not days; fog is mostly in winter. To be replaced by HungaroMet data.
Context: stopping distance		
speed	130 km/h	Hungarian motorway speed limit.
latency, braking	0.5 s, 6 m/s^2	system reaction; firm but not full braking (dry road allows 8–9 m/s^2). $d = vt + v^2/2a \approx 127$ m, so 250 m leaves 68 frames.

Datasheet PDF: `https://levelfivesupplies.com/wp-content/uploads/2023/08/InnovizTwo-GMSL-Series-v3.1-1.pdf`

6 Main assumptions

Flat rectangular targets facing the lidar; beams are points (no partial hits); beams fail independently; the shape of $p(R)$ beyond the datasheet point follows from the Poisson model (another common shape, the speckle model, gives 248 m instead of 235 m for the pedestrian, so ours is the cautious one); false positives (1 % per pixel) are ignored; one frame is judged on its own.